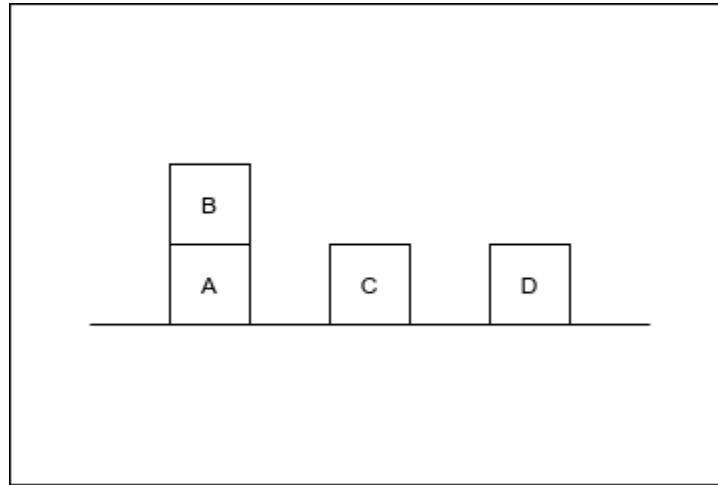


Blocks World Problem

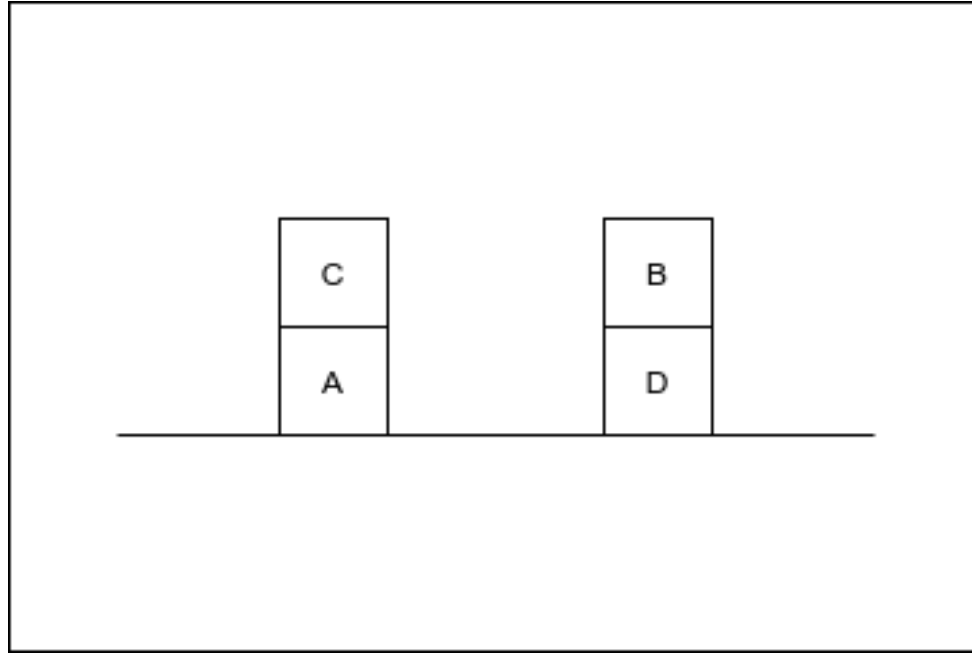
Assume the following predicates:

- **ON(A,B)** : Block A is on B
- **ONTABLE(A)** : A is on table
- **CLEAR(A)** : Nothing is on top of A
- **HOLDING(A)** : Robot Arm is holding A
- **ARMEMPTY** : Robot Arm is empty



Initial State Description:

$\text{ON}(\text{B}, \text{A}) \wedge \text{ONTABLE}(\text{A}) \wedge \text{ONTABLE}(\text{C}) \wedge \text{ONTABLE}(\text{D}) \wedge$
 $\text{CLEAR}(\text{B}) \wedge \text{CLEAR}(\text{C}) \wedge \text{CLEAR}(\text{D}) \wedge \text{ARMEMPTY}$



Goal State — $\text{ON}(\text{C}, \text{A}) \wedge \text{ON}(\text{B}, \text{D}) \wedge \text{ONTABLE}(\text{A}) \wedge \text{ONTABLE}(\text{D}) \wedge$
 $\text{CLEAR}(\text{B}) \wedge \text{CLEAR}(\text{C}) \wedge \text{ARMEMPTY}$

Assume that the Robot Arm can perform 4 operations:

1. **STACK(X,Y)** : Stacking Block X on Block Y
2. **UNSTACK(X,Y)** : Picking up Block X which is on top of Block Y
3. **PICKUP(X)** : Picking up Block X which is on top of the table
4. **PUTDOWN(X)** : Put Block X on the table

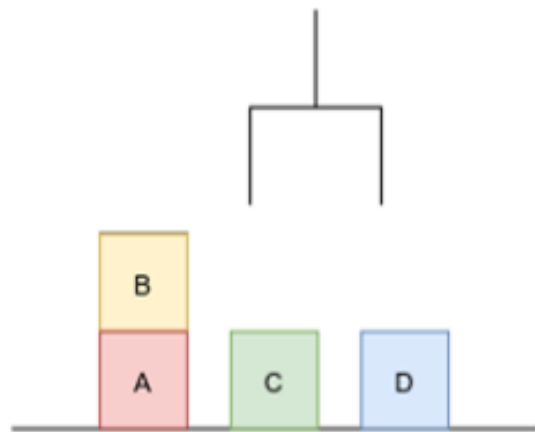
All the four operations that a robot can perform have certain preconditions represented in the form of predicates (as given below)

OPERATORS	PRECONDITION	DELETE	ADD
STACK(X,Y)	CLEAR(Y) $\wedge$ HOLDING(X)	CLEAR(Y) HOLDING(X)	ARMEMPTY ON(X,Y)
UNSTACK(X,Y)	ARMEMPTY $\wedge$ ON(X,Y) $\wedge$ CLEAR(X)	ARMEMPTY $\wedge$ ON(X,Y)	HOLDING(X) $\wedge$ CLEAR(Y)
PICKUP(X)	CLEAR(X) $\wedge$ ONTABLE(X) $\wedge$ ARMEMPTY	ONTABLE(X) $\wedge$ ARMEMPTY	HOLDING(X)
PUTDOWN(X)	HOLDING(X)	HOLDING(X)	ONTABLE(X) $\wedge$ ARMEMPTY

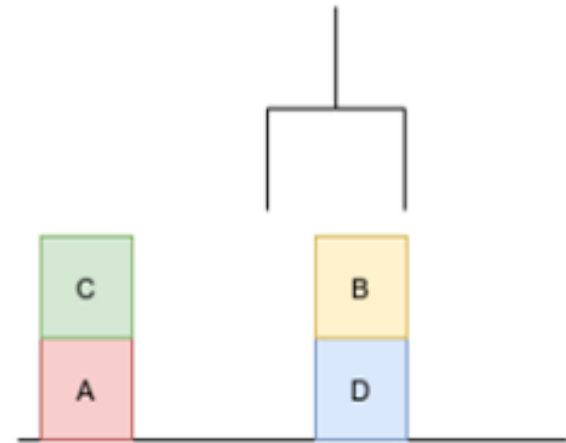
The effect of these operations is represented using two lists **ADD** and **DELETE**. DELETE List contains the predicates which will cease to be true once the operation is performed. ADD List on the other hand contains the predicates which will become true once the operation is performed.

The problem is to build a plan to move from the Initial state to the Goal state as shown below.

Write the Plan (sequence of operations) along with the preconditions and the contents of ADD List & DELETE List.



Initial State



Goal State